SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

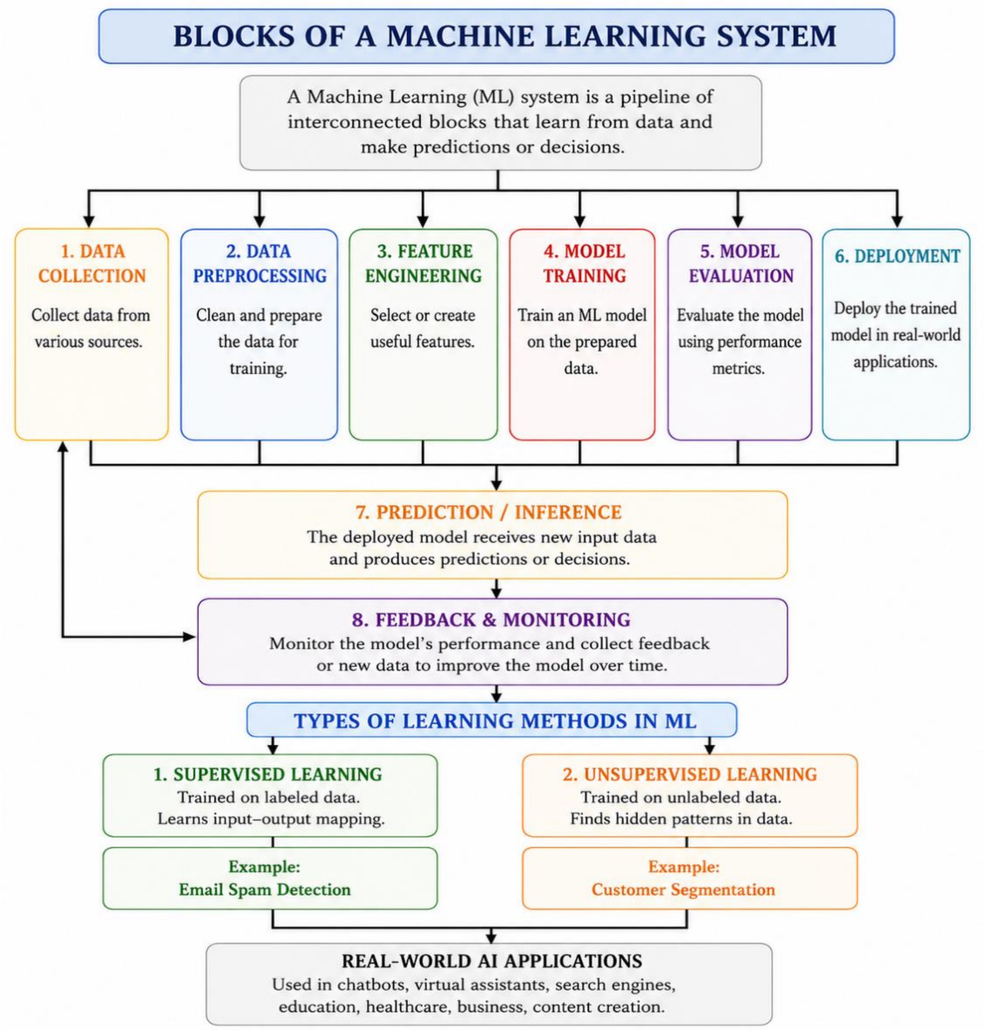
CONCEPT MAPPING

Code of Conduct:

I **Chalamala Sai Harshitha (192424346)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

1. **With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.**



Supervised Learning:

Problem: Email Spam Detection

- Input: Email text
- Output: Spam / Not Spam
- Training data contains labels.
- Algorithms: Naïve Bayes, Decision Tree, Logistic Regression.

Example

Email	Label
Win ₹10,000 now!	Spam
Meeting at 2 PM	Not Spam

Unsupervised Learning:

Problem: Customer Segmentation

- Customer data has no labels.
- The algorithm groups similar customers automatically.
- Algorithm: K-Means Clustering.

Example

Customer Data

│

▼

K-Means Clustering

├── Group 1 (Students)

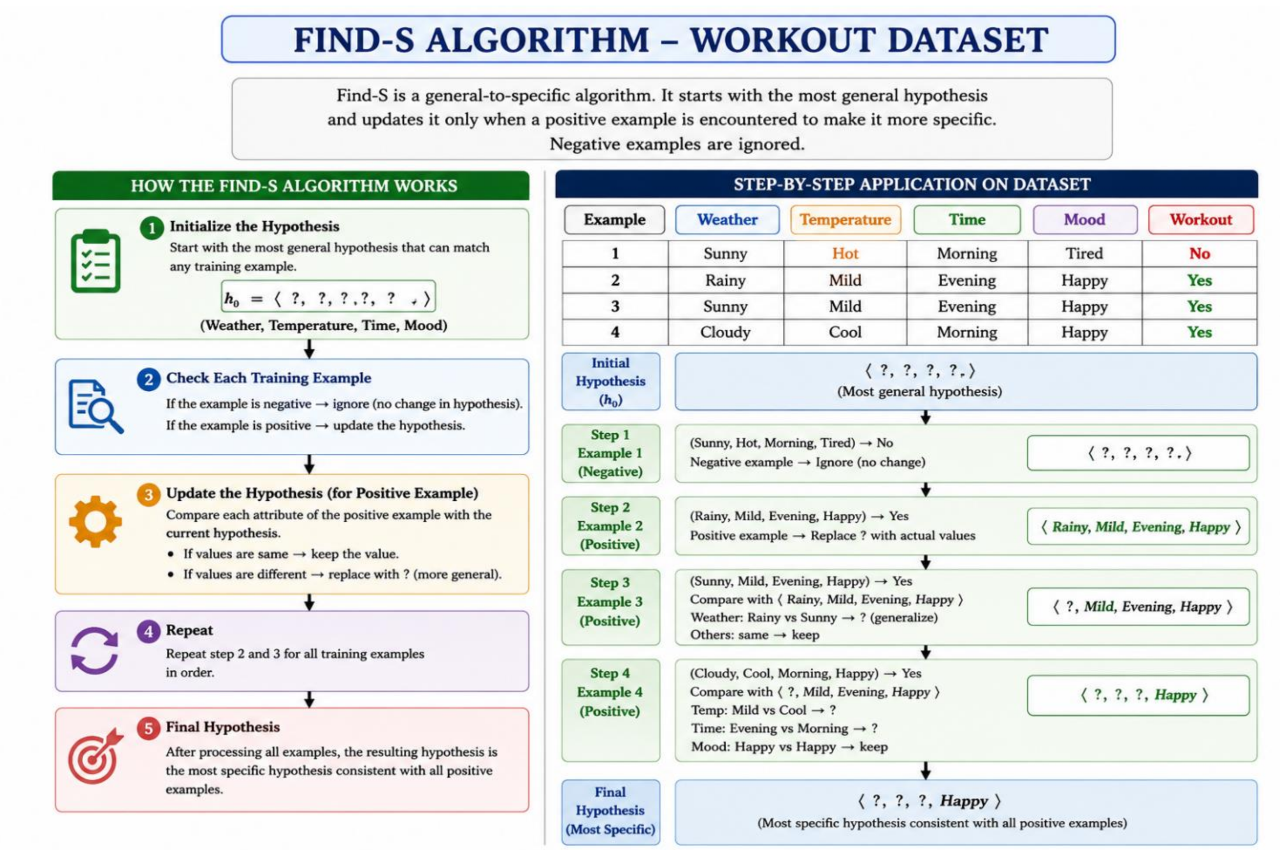
├── Group 2 (Working Professionals)

└── Group 3 (Senior Citizens)

2. **In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only.**

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes



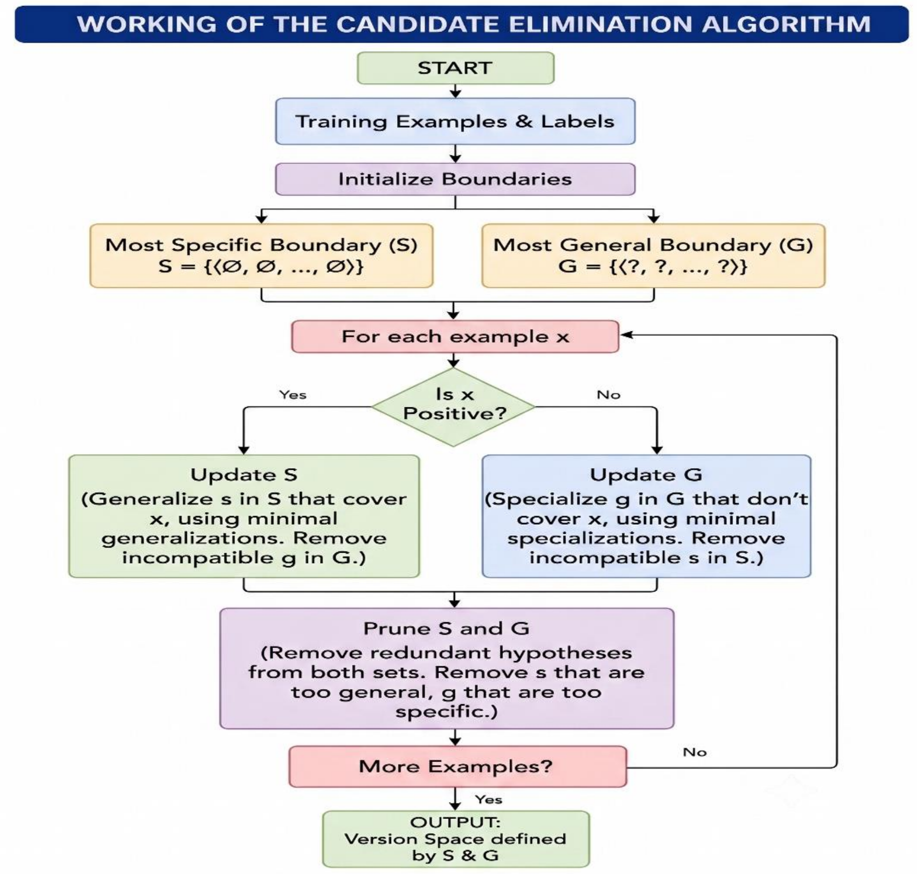
Conclusion

The Find-S algorithm considers only positive examples and generalizes differing attributes. Therefore, the maximally specific hypothesis is:

<?, ?, ?, Happy>

3. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example.
Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No



Initialization

S0 = <Ø, Ø, Ø, Ø>

G0 = <?, ?, ?, ?>

Example 1 (Positive)

S1 = <Young, High, Frequent, Mobile>

G1 = <?, ?, ?, ?>

Example 2 (Positive)

Internet Usage differs

S2 = <Young, High, ?, Mobile>

G2 = <?, ?, ?, ?>

Example 3 (Negative)
Specialize G
S3 = <Young, High, ?, Mobile>
G3 =
<Young, ?, ?, ?>
<?, High, ?, ?>
<?, ?, ?, Mobile>

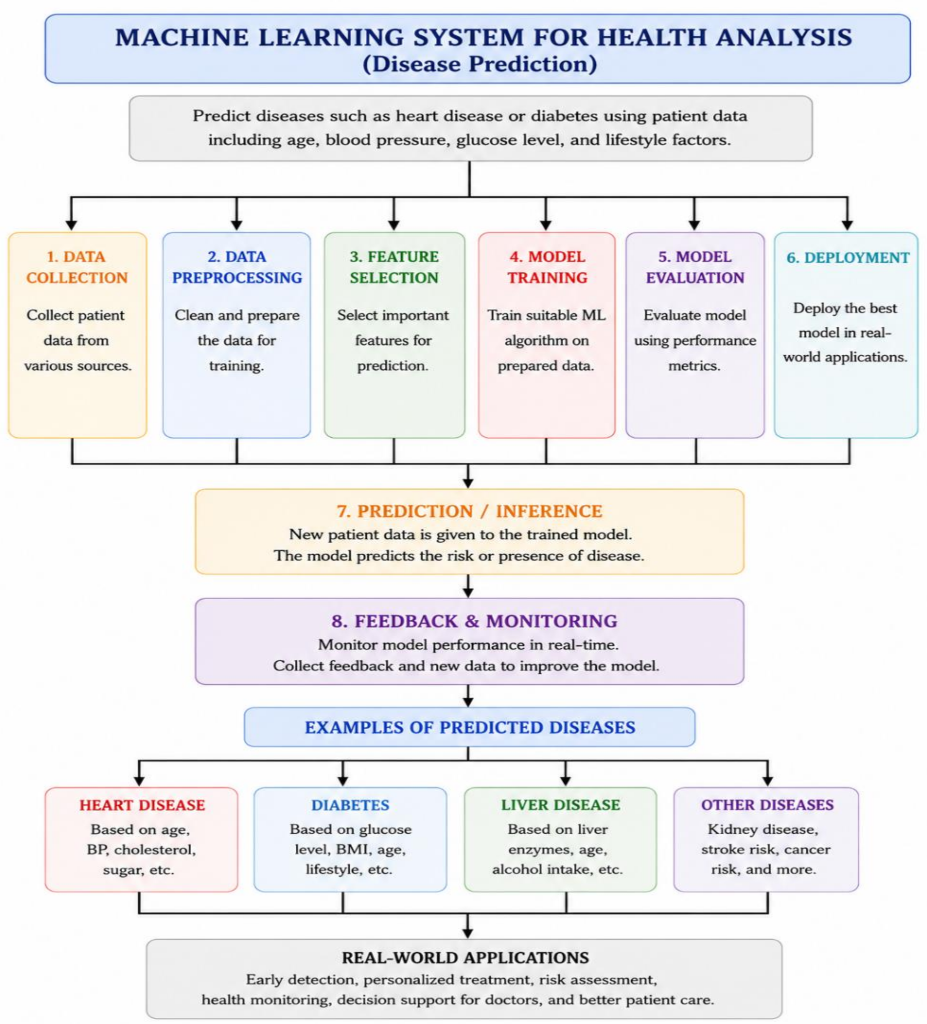
Example 4 (Positive)
Income differs
S4 = <Young, ?, ?, Mobile>
G4 =
<Young, ?, ?, ?>
<?, ?, ?, Mobile>

Example 5 (Negative)
The example is already excluded by the current boundaries.

Final Version Space
Specific Boundary (S): <Young, ?, ?, Mobile>
General Boundary (G): <Young, ?, ?, ?>
<?, ?, ?, Mobile>

Conclusion
The Candidate Elimination algorithm updates both the **Specific (S)** and **General (G)** boundaries after every training example. The final version space indicates that customers who are **Young** and/or use a **Mobile device** are most likely to subscribe to the service.

4. **Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.**

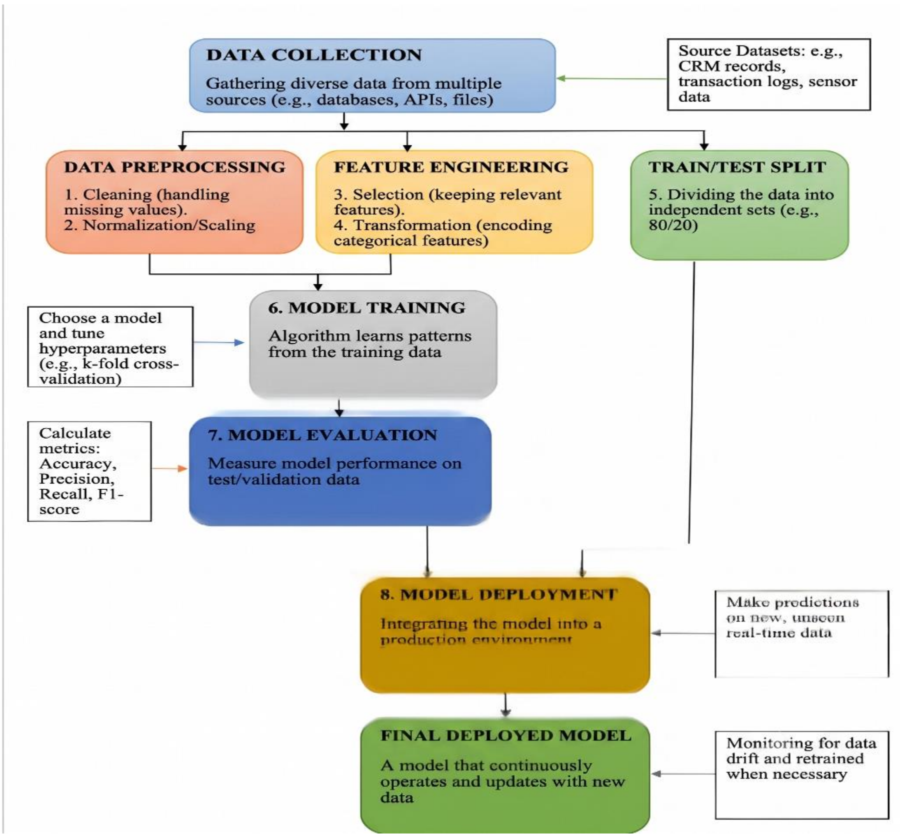


Algorithm	**Justification**
Random Forest	High accuracy, reduces overfitting, handles multiple health features, and works well with large datasets.
Logistic Regression	Suitable for binary classification such as predicting Disease: Yes/No. Simple and easy to interpret.
Decision Tree	Easy to understand, creates clear decision rules, and handles both numerical and categorical data effectively.

Conclusion

A Random Forest-based model accurately predicts diseases such as diabetes and heart disease, enabling early diagnosis and better healthcare decisions.

5. **Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.**



Stages of Machine Learning Pipeline:

Stage	**Description**
Data Collection	Collects data from databases, sensors, websites, or files.
Data Preprocessing	Removes missing values, duplicates, and noise.
Feature Engineering	Selects or creates useful features for better prediction.
Model Training	Learns patterns from training data using ML algorithms.
Model Evaluation	Measures performance using Accuracy, Precision, Recall, and F1-Score.
Model Deployment	Deploys the trained model for real-time predictions.

Training Error vs Generalization Error

Training Error	**Generalization Error**
Error on training data.	Error on unseen test data.
Usually, low.	Indicates real-world performance.
Used during model training.	Used to measure model reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand